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 Studierichting: Informatica
 Oefeningenreeks 3F nr. 1.3

$$f(x) = \frac{x^3 + 2x^2 + 2x - 2}{x^2 + 1}$$

$$D = 0^2 - 4 \cdot 1 \cdot 1 < 0, \text{ dus geen VA.}$$

$$\text{HA: } \lim_{x \rightarrow \infty} \frac{x^3 + 2x^2 + 2x - 2}{x^2 + 1} = \frac{\infty}{\infty}$$

$$\stackrel{\text{H}}{=} \lim_{x \rightarrow \infty} \frac{3x^2 + 4x + 2}{2x} = \frac{\infty}{\infty}$$

$$\stackrel{\text{H}}{=} \lim_{x \rightarrow \infty} \frac{6x + 4}{2} = \infty, \text{ dus geen HA.}$$

$$\text{SA: } \lim_{x \rightarrow \infty} \frac{\frac{x^3 + 2x^2 + 2x - 2}{x^2 + 1}}{x} = \lim_{x \rightarrow \infty} \frac{x^3 + 2x^2 + 2x - 2}{x^3 + x} = \frac{\infty}{\infty}$$

$$\stackrel{\text{H}}{=} \lim_{x \rightarrow \infty} \frac{3x^2 + 4x + 2}{3x^2 + 1} = \frac{\infty}{\infty}$$

$$\stackrel{\text{H}}{=} \lim_{x \rightarrow \infty} \frac{6x + 4}{6x} = \frac{\infty}{\infty}$$

$$\stackrel{\text{H}}{=} \lim_{x \rightarrow \infty} \frac{6}{6} = 1 = a$$

$$b = \lim_{x \rightarrow \infty} \frac{x^3 + 2x^2 + 2x - 2}{x^2 + 1} - x$$

$$= \lim_{x \rightarrow \infty} \frac{x^3 + 2x^2 + 2x - 2 - x^3 - x}{x^2 + 1}$$

$$= \lim_{x \rightarrow \infty} \frac{2x^2 + x - 2}{x^2 + 1} = \frac{\infty}{\infty}$$

$$\stackrel{\text{H}}{=} \lim_{x \rightarrow \infty} \frac{4x + 1}{2x} = \frac{\infty}{\infty}$$

$$\stackrel{\text{H}}{=} \lim_{x \rightarrow \infty} \frac{4}{2} = 2$$

$$SA = x + 2$$

Ligging tegenover de schuine asymptoot:

$$\begin{aligned} & \frac{x^3 + 2x^2 + 2x - 2}{x^2 + 1} - x + 2 \\ &= \frac{x^3 + 2x^2 + 2x - 2 - (x + 2)(x^2 + 1)}{x^2 + 1} \\ &= \frac{x^3 + 2x^2 + 2x - 2 - x^3 - x - 2x^2 - 2}{x^2 + 1} = \frac{x - 4}{x^2 + 1} \end{aligned}$$

x	$-\infty$	4	$+\infty$
$\frac{x-4}{x^2+1}$		$- \quad 0 \quad +$	

References